

Unit-5 Risk Management [5 Hrs.]

Risk Identification, Planning, Evaluation and Management,
Categories of Risk,
Framework for dealing with risk,
Evaluating Risks to the schedule.

This unit focuses on the critical discipline of risk management in software project management.

Risk management is the process of identifying, assessing, mitigating, and monitoring risks that can impact a project's success.

This unit provides a comprehensive understanding of how to effectively manage risks throughout the project's lifecycle.

Risk Management

- ◆ Risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives such as scope, schedule, cost, and quality.
- ◆ Risk Management is the process of identifying and migrating risk.
- ◆ Risk management is defined as “the systematic application of management practices, policies, and procedures for identifying, analyzing, controlling and monitoring risk.

When developing risk management framework at least 5 critical components must be considered:

- Risk identification,
- Risk measurement and assessment,
- Risk mitigation (Minimization),
- Risk reporting and monitoring,
- Risk governance

Why is it important?

- ◆ Risk affects all aspects of your project-your budget, your schedule, your scope, the agreed level of quality, and so on.
- ◆ Increase probability of positive event.
- ◆ Reduce the occurrence of negative events.

Risk management planning

- ◆ Analysis and decision making to implement risk management according to size and complexity of the project.
- ◆ Understand the general level of risk your project faces
- ◆ Stakeholders will be involved in planning risk management.
- ◆ Risk Management Plan should include:
 - Schedule Risk
 - Cost Risk
 - Quality Risk
 - Scope Risk
 - Resource Risk
 - Customer Satisfaction Risk

Risk Planning, Evaluation, and Management:

Risk planning, evaluation, and management are integral components of the risk management process in software project management. These processes help ensure that potential risks are identified, assessed, and effectively addressed.

Here's an overview of each aspect:

Risk Planning:

Definition: Risk planning involves developing a comprehensive plan for how risks will be managed throughout the project lifecycle.

Risk Evaluation:

Definition: Risk evaluation is the process of assessing the identified risks to determine their potential impact and likelihood.

Risk Management:

Definition: Risk management involves the development and execution of strategies to mitigate, respond to, or accept identified risks.

Risk Planning:

Risk planning involves developing a comprehensive plan for how risks will be managed throughout the project lifecycle.

Key Steps:

- **Risk Management Plan:** Create a formal risk management plan that outlines the strategy for risk identification, assessment, mitigation, and monitoring.
- **Roles and Responsibilities:** Clearly define roles and responsibilities of team members and stakeholders involved in risk management.
- **Risk Thresholds:** Set risk thresholds to determine when risks should trigger specific actions or mitigation measures.
- **Communication Plan:** Develop a communication plan to ensure that all stakeholders are informed about the project's risk management process.
- **Documentation:** Establish a system for documenting identified risks, their assessments, and planned responses.

Risk Evaluation:

Definition: Risk evaluation is the process of assessing the identified risks to determine their potential impact and likelihood.

Key Steps:

- **Risk Assessment:** Assess each risk by considering its potential impact on project objectives and the likelihood of its occurrence.
- **Risk Prioritization:** Prioritize risks based on their assessed significance. High-priority risks require immediate attention and detailed mitigation plans.
- **Quantitative and Qualitative Analysis:** Utilize quantitative techniques (e.g., risk matrices) and qualitative analysis to evaluate risks from multiple perspectives.
- **Scenario Analysis:** Consider different scenarios in which risks may occur and assess their impact under various conditions.

Risk Management:

Definition: Risk management involves the development and execution of strategies to mitigate, respond to, or accept identified risks.

Key Steps:

- **Risk Mitigation Strategies:** Develop specific strategies for mitigating risks, including actions to reduce the probability of occurrence and minimize potential impacts.
- **Risk Response Plans:** Create response plans for each risk, outlining actions to take if the risk materializes.
- **Risk Monitoring:** Continuously monitor identified risks throughout the project to track changes in their likelihood and impact.
- **Contingency Planning:** Develop contingency plans for high-priority risks to have predefined actions in place if they occur.

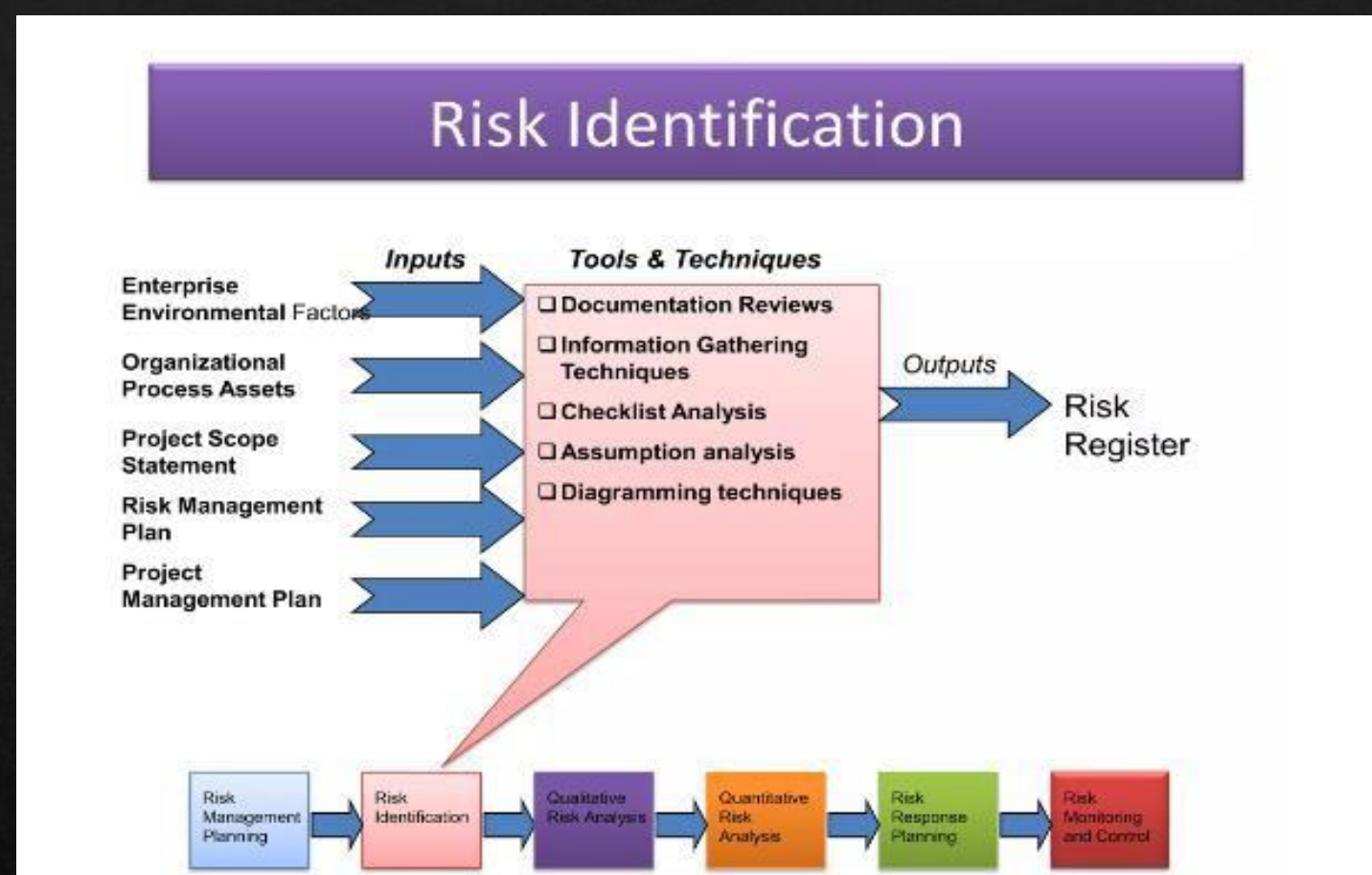
How do we manage risk ?

Framework for Dealing with Risk

- ◆ When developing risk management framework at least 5 critical components must be considered:
 - Risk identification,
 - Risk measurement and assessment,
 - Risk mitigation (Minimization),
 - Risk reporting and monitoring,
 - Risk governance

1. Risk Identification

- ◆ Identify individual risks that might affect your project by making a list
- ◆ Which risk has more probability of affecting the project?
- ◆ Use wide variety of inputs, tools , and technique to identify all the risks to the project.
- ◆ Performed → Throughout the life of the project



Definition: The first step in the framework involves identifying potential risks. This is done through techniques like brainstorming, checklists, and expert interviews.

Activities: Project stakeholders collaborate to generate a comprehensive list of risks that could impact the project's success. The goal is to capture a wide range of potential risks.

Techniques and Methods for Identifying Potential Risks:

Brainstorming: Encourage team members and stakeholders to brainstorm and list potential risks. This collaborative approach generates diverse perspectives.

Checklists: Utilize risk checklists that contain common risk categories and examples. These checklists serve as a structured starting point for risk identification.

Root Cause Analysis: Investigate past project failures and issues to identify root causes. Addressing these underlying issues can prevent their recurrence in future projects.

SWOT Analysis: Analyze the project's strengths, weaknesses, opportunities, and threats (SWOT). The "T" represents potential threats and risks.

Expert Interviews: Consult subject matter experts who have experience in similar projects. Their insights can uncover hidden risks.

Historical Data: Review historical project data and lessons learned reports to identify recurring risks that have impacted previous projects.

The Role of Stakeholders in Identifying and Communicating Risks:

Stakeholders play a crucial role in the risk identification process.

Their active participation is essential for comprehensive risk assessment.

Project Team: Team members with various roles and responsibilities may have unique insights into potential risks related to their areas of expertise.

Clients and End Users: Clients and end users can identify risks related to project requirements, functionality, and usability that might not be apparent to the project team.

Management and Leadership: Project managers and leaders often have experience in risk management and can provide valuable input on potential risks.

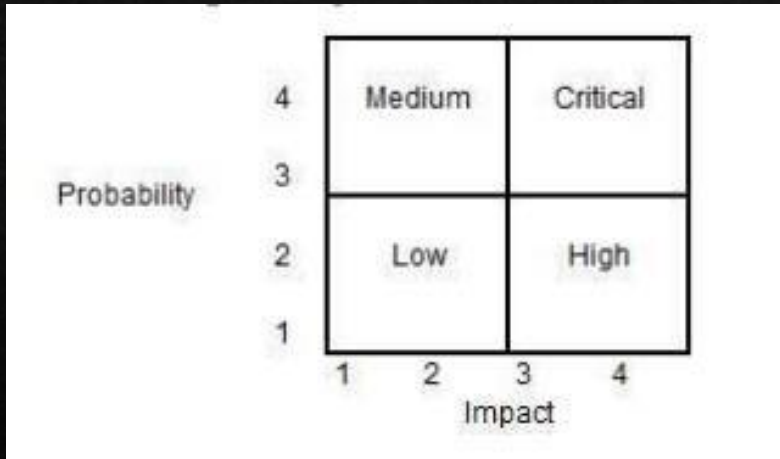
External Experts: Depending on the project's nature, external experts, consultants, or industry specialists can contribute insights into specific risks.

Open Communication: Encourage an environment of open communication where stakeholders feel comfortable expressing their concerns and observations regarding risks.

Documentation: Maintain clear records of identified risks and their sources, including stakeholders. This documentation is essential for reference and tracking throughout the project.

2. Risk Measurement & Assessment (Analysis)

- ◇ begin putting this list of risks in a **risk register**—a chart that lays out each risk, followed by information like priority level and mitigation plans
- ◇ Helps to identify and manage potential problem
- ◇ Using the matrix, the project manager can categorize the risk into four categories as Low, Medium, High and Critical.
- ◇ The probability of occurrence and the impact on the project are the two parameters used for placing the risk in the matrix categories.



Probability	4	Medium	Critical
	3		
	2	Low	High
	1		
		1	2
		3	4
		Impact	

Definition: After identifying risks, the next step is to assess them. This involves evaluating the probability and impact of each risk on the project's objectives.

Activities: Project teams use both qualitative and quantitative methods to assess risks. This can include risk matrices, probability and impact assessments, and scenario analysis.

- ◇ The categorization of risk can be done on the percentage basis , very low(75%) ,here percentage resemble either its probability of occurrence in a project or its impact on the project.

3. Risk Mitigation (Risk Response planning)

- ◇ assigning each risk a high, medium, or low priority based on the factors you've determined
- ◇ Eliminate threats before they happen.
- ◇ Decrease impact of threat
- ◇ Contingency Plan (Do Something if risk happens)
- ◇ Fallback Plan (DO something is contingency plans are not effective)

Definition: Once risks are assessed, response plans are developed. These plans outline strategies for dealing with each risk, including risk mitigation, risk acceptance, risk avoidance, and risk transfer.

Activities: For high-priority risks, detailed response plans are created. This includes specifying actions to reduce the likelihood and impact of the risk or defining what steps to take if the risk materializes.

Negative Risk or Threats	Positive risk or Threats
Avoid Transfer Mitigate (minimize) Accept	Exploit Share enhance

4. Risk Monitoring & reporting

- ◇ set up a process to monitor each risk as your project begins
- ◇ Risk Assessment → Schedule regularly to identify new risk
- ◇ Risk Audit → Examine the effectiveness of planned risk response
- ◇ Trend Analysis → Monitor overall project performance

Definition: Risk monitoring is an ongoing process that involves tracking identified risks throughout the project's lifecycle.

Activities: Project teams continuously review the status of identified risks, assess any changes in their likelihood or impact, and update risk response plans as necessary. This ensures that the project remains proactive in addressing evolving risks.

5. Risk governance

- ◆ Implement all of the risk management steps defined above.
- ◆ Risk governance involves defining the roles of all employees, segregating duties, and assigning authority to individuals, committees, and the board for approval of core risks, risk limits, exceptions to limits, and risk reports, and also for general oversight.

Risk Management Framework Steps

Step	Description
Prepare	➤ focuses on getting the organization ready to adopt a formalized risk management strategy
Categorize	➤ organizations begin assessing the risks that have been identified. ➤ This may mean assessing the impact of the various risks and prioritizing the risks that need to be addressed.
Select	➤ involves <u>choosing the controls</u> that will be used to protect affected systems to minimize or mitigate the risks that have been identified.
Implement	➤ This is where the selected controls are put into place in an effort to head off risks that might exist.
Assess	➤ It seeks to determine whether the selected controls were implemented correctly and if those controls are delivering the desired result ➤ This means making sure any mechanisms that have been implemented are reducing risks in a quantifiable way without accidentally introducing new risks in the process.
Authorize	➤ the authorize phase is more of an overview by senior members of the organization who are looking to make sure that risk mitigation strategies are working

What are the benefits?

- ◆ Effective use of resources
- ◆ Promoting continuous improvement
- ◆ Fewer Shocks and failures
- ◆ Strategic business planning
- ◆ Raised awareness of significant risks.
- ◆ Quick grasp of new opportunity
- ◆ Enhancing Communication
- ◆ Reassurance stakeholders
- ◆ Recognition of responsibility

Categories of Risk

In software project management, risks can be categorized into several distinct categories based on their nature and potential impact on the project. Understanding these categories is essential for comprehensive risk assessment and management.

Here are the primary categories of risk:

1. Technical Risks:

2. Business Risks:

3. Operational Risks:

4. External Risks:

5. Project Management Risks:

6. Environmental Risks:

1. Technical Risks:

- ◇ Threaten the quality and timeliness of the software to be produced.
- ◇ If the technical risk become a reality , implementation may become difficult or impossible
- ◇ Some reasons for Technical risks –
 - ◇ Frequent changes in requirement
 - ◇ Less use of future technologies
 - ◇ Less number of skilled employee
 - ◇ High complexity in implementation
 - ◇ Improper integration of modules

Definition: Technical risks are associated with challenges related to the development and implementation of software. These risks can affect the project's ability to deliver a working solution.

Examples: Technical risks can include issues with software architecture, coding errors, performance bottlenecks, integration challenges, and technology dependencies.

2. Business Risks:

- ◊ Business risks are uncertain factors, internal or external, that threaten the financial health of an organization.
- ◊ Examples of external business risks would be natural disasters or cyber attacks. Internal business risks are threats that come from within the company, such as falling out of compliance, having too much debt, or labor disputes.
- ◊ Top business risk are :
 - **Market risk:** Building an excellent product or system that no one really wants.
 - **Sales risk:** Building a product that the sales force does not understand how to sell.
 - **Management Risk:** losing the support of senior management due to change in focus or change

Definition: Business risks pertain to factors that can impact the project's alignment with business goals and objectives. They often revolve around financial and strategic aspects.

Examples: Business risks can encompass market fluctuations, changes in project scope, financial constraints, evolving business priorities, and competition.

3. Operational Risks:

Definition: Operational risks concern the day-to-day operational aspects of the project and its impact on project success.

Examples: Operational risks may involve resource shortages, team dynamics, communication breakdowns, schedule deviations, and process inefficiencies.

4. External Risks:

Definition: External risks are related to factors that are beyond the direct control of the project team. These risks are influenced by external entities or events.

Examples: External risks can encompass changes in regulatory requirements, legal issues, geopolitical events, market dynamics, and vendor-related challenges.

5. Project Management Risks:

Definition: Project management risks are specific to the project's planning and execution. They relate to challenges that may arise during the project management process.

Examples: Project management risks may include inaccurate project estimation, scope creep, resource allocation issues, and ineffective communication.

6. Environmental Risks:

Definition: Environmental risks are connected to factors in the project's environment that could impact its success. These risks are often related to the project's context.

Examples: Environmental risks may involve issues with external infrastructure, cultural differences, remote work challenges, and natural disasters.

Evaluating risk to the schedule

- ◇ not all risks can be eliminated - even those that are classified as avoidable or
- ◇ By identifying and categorizing those risks, and in particular, their likely effects on the duration of planned activities, we can assess what impact they are likely to have on our activity plan.
- ◇ Two methods for assessing the effects of these uncertainties on the project schedule
 - ◇ Risk Matrix

Severity

- ◆ **Severity** is the amount of damage or harm a hazard could create and it is often ranked on a four point scale as follows:
 - ◆ **Catastrophic - 4:** Operating conditions are such that human error, environment, design deficiencies, element, subsystem or component failure, or procedural deficiencies may commonly cause death or major system loss, thereby requiring immediate cessation of the unsafe activity or operation.
 - ◆ **Critical - 3:** Operating conditions are such that human error, environment, design deficiencies, element, subsystem or component failure, or procedural deficiencies may commonly cause severe injury or illness or major system damage thereby requiring immediate corrective action.
 - ◆ **Marginal - 2:** Operating conditions may commonly cause minor injury or illness or minor systems damage such that human error, environment, design deficiencies, subsystem or component failure, or procedural deficiencies can be counteracted or controlled without severe injury, illness, or major system damage.
 - ◆ **Negligible - 1** Operating conditions are such that personnel error, environment, design deficiencies, subsystem or component failure, or procedural deficiencies will result in no, or less than minor, illness, injury, or system damage.

Probability

- ◇ **Probability** is the likelihood of the hazard occurring and it is often ranked on a five point scale:
 - ◇ **Frequent - 5:** Likely to occur often in the life of an item
 - ◇ **Probable - 4:** Will occur several times in the life of an item
 - ◇ **Occasional - 3:** Likely to occur sometime in the life of an item.
 - ◇ **Remote - 2:** Unlikely but possible to occur in the life of an item.
 - ◇ **Improbable - 1:** So unlikely, it can be assumed an occurrence may not be experienced.

How to Use the Risk Assessment?

- ◇ The Risk Assessment values are determined by multiplying the scores for the Probability and Severity values together.
- ◇ The higher the risk assessment, the greater the overall risk for the project.
- ◇ This method helps balance the weight of severity and probability

Probability

Severity

	Catastrophic: 4	Critical: 3	Moderate: 2	Marginal: 1
Frequent: 5	High - 20	High - 15	High - 10	Medium - 5
Probable: 4	High - 16	High - 12	Serious - 8	Medium - 4
Occasional: 3	High - 12	Serious - 9	Medium - 6	Low - 3
Remote: 2	Serious - 8	Medium - 6	Medium - 4	Low - 2
Improbable: 1	Medium - 4	Low - 3	Low - 2	Low - 1

Evaluating Risks to the Project Schedule

Evaluating risks to the project schedule is a crucial aspect of risk management in software project management. This process involves assessing potential risks that could impact the project's timeline and developing strategies to mitigate their effects. Here's a breakdown of how this evaluation is conducted:

- 1. Risk Identification:** Begin by identifying risks that have the potential to affect the project schedule. These risks can come from various sources, including technical, business, or operational aspects.
- 2. Risk Assessment:** Assess the identified risks to understand their potential impact on the project schedule. This step involves evaluating the likelihood and severity of schedule delays.
- 3. Prioritization and Classification:** Prioritize the assessed risks based on their significance and potential impact on the project schedule. Risks can be classified into categories like high, medium, or low priority.
- 4. Risk Response Planning:** Develop specific risk response plans for high-priority risks that could significantly impact the project schedule. These plans outline actions to take in case the risk materializes.
- 5. Contingency Planning:** Develop contingency plans for high-priority risks. Contingency plans outline predefined actions to be executed if a particular risk leads to schedule delays.
- 6. Continuous Monitoring:** Continuously monitor the identified schedule risks throughout the project's lifecycle to track any changes in their likelihood and impact.
- 7. Integration with Project Management:** Integrate the risk assessment and response activities with the overall project management process. This alignment ensures that the project schedule remains a central focus.